GENERATIVE AI: PRACTICAL SESSION

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Environment

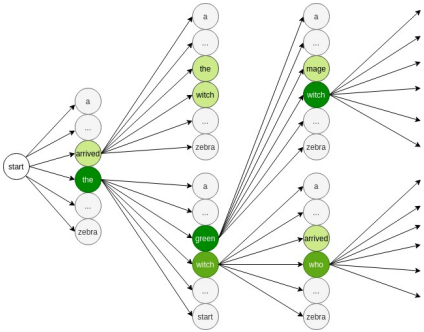
1. To play quickly with LLM: `https://textsynth.com/completion.html`
   - Distinguish LLM with/wo instruction
   - Understand temperature
2. Resolve real use case with real LLMs (will require an -free- account)
   - chatGPT : `https://chatgpt.com`
   - Claude : `https://claude.ai/new`
   - Google Gemini : `https://gemini.google.com/app`
   - Mistral : `https://chat.mistral.ai/chat`
   - Llama 3 : `https://huggingface.co/chat/`
   - For image : `https://stablediffusionweb.com/fr/app/image-generator`
3. Play with local LLM : ollama... But $>$ 4Gb per model
4. Benchmark (really difficult task $\Rightarrow$ human !)
   - `https://chat.lmsys.org`

Parametrization

- Choice of the LLM
  - Which size for which application?
  - Which budget?
  - Which ethical constraint?
  - Big + general + expensive *vs* small + adaptation + cheap
- Temperature

Parametrization

- Choice of the LLM
- Temperature



- Next word = sampling in a distribution
- Computation:
  word scores (logit) $ws\mathbb{R} \Rightarrow$ distrib. $p(w)$

$$p(w) = \frac{\exp(-\alpha T ws)}{\sum_i \exp(-\alpha T ws_i)}$$

- $T > 1 \Rightarrow$ Uniform = diversity
- $T < 1 \Rightarrow$ Dirac = Frozen/more deterministic

Getting started with LLM

`https://textsynth.com/completion.html`

- Distinguish LLM with/wo instruction
- Understand temperature

Compare instruct model *vs* pre-trained on a task, play with the temperature

What would be the story of a frog going in space?

Prompt & Information Access

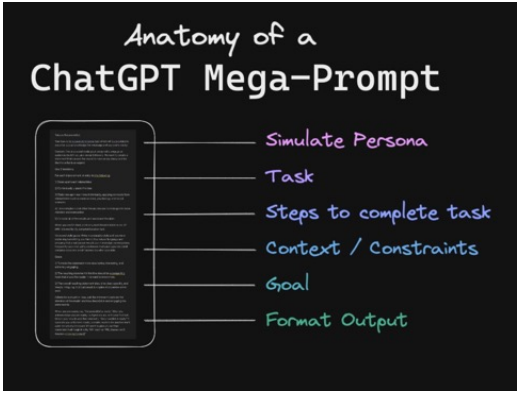
Relevant information = long prompt

Ranked by importance:

1. What is the question/task?
2. Who ask? / What is the context?
3. Steps to carry out the task
4. Output format / style
5. Exemple of pairs (questions/answers)

Solution 1: long prompt
Solution 2: Dialogue, ask several questions

Ordered when writing:



https://chatgptprompts.guru/what-makes-a-good-chatgpt-prompt/

Example (1) : Classical information access

- What is the definition of a *prompt*?

  In the field of LLMs, quickly explained with a simple vocabulary

- Try to write 5 lines about a domain in which you are not a specialist. Proposition:
  - What is *Conformal prediction*?
  - Differences between *Rationalism* and *Empiricism* in Enlightenment philosophy
  - Differences between a *metaphor* and an *allegory*?
  - What are the mainstays of the Mongolian economy?

- Let's ask the same question looking for a detailed answer for a specialist

Which risks? Which use? Which confidence?

Example (2) : NLP task

Given the sentence (from CONLL03):

`The European Commission said on Thursday it disagreed with German advice to consumers to shun British lamb until scientists determine whether mad cow disease can be transmitted to sheep.`

1. Extract the following entities with their types :*(place, person, organisation, date)*
2. Format the output in JSON
   - Is the result the same with formattting constraints?
3. Try some prompt from the appendix of GPTNER: `https://arxiv.org/pdf/2305.15444`
4. API Usage: `https://cookbook.openai.com/`

```
{
  "entities": [
    {
      "type": "Organization",
      "value": "European Commission"
    },
...
```

Creativity, formatting & rephrasing

Brainstorming

- Build a detailed semester schedule for an advanced course in your field:
  - $\approx$ 10 courses
  - 2-3 parts in each courses
  - 5 to 10 bullet points in every part

- Obviously, you could do it: takes time + risk of forgetting things

$\Rightarrow$ Try to do it with LLMs

Motivation/recommendation letter

- Write a motivation letter for an MSc. in your field... And imagine your are undergraduate
  - Which keywords could you give at first? Which one could you discover?
  - Check your estimated strength wrt the targeted MSc.
  - Evaluate the generated letter as an examiner
- Write a recommendation letter for a student applying for a PhD in your field
  - How long do you know him? The course in which you met him? Main qualities...
  - + subject of the PhD

Meeting minutes

- Take some (non confidential) notes from a recent meeting

or write quickly some notes

or use this file:

1. Make the LLM write the meeting report

Summary

1. Generate a summary about the wikipedia page: `https://en.wikipedia.org/wiki/Variational_Bayesian_methods`
2. 10 lines, non technical vocabulary
3. 50 lines, more technical

4. Same thing with a recent news (for instance) `https://edition.cnn.com/2024/07/09/politics/biden-nato-analysis/index.html`

Text (article?) improvement

1. Get an article (your last one, your advisor's latest one...)
2. Copy-paste on paragraph
3. Add some noise (spelling mistake, bad formulation...)
4. Try to improve (optionnally translate)

⇒ What do you think about the result?

- Cheating?
- Simply exploiting new tools?
- Social rebalancing?

Conclusion

- LLM are not providing specific information
  - You have to give them to the system
  - Check what comes out first: these commonplace things are the same for everyone.
    - Keep the overall layout,
    - Personalize and reinforce qualities
- Ask the LLM for the list of qualities (& weaknesses) for a specific position

Is it cheating to use LLM for application/founding projects?

LLM are hard to detect... But it is easy to be caught!

See Guillaume Cabanac work on *Problematic Paper Screener*:
`https://www.irit.fr/~Guillaume.Cabanac/`

- Prompt forgetting :

#1 Statcheck comment accepted September 2016

Using the R package statcheck (v1.0.1), the HTML version of this article was scanned on 2016-08-05 for statistical results (t, r, F, Chi2, and Z values) reported in APA format (for specifics, see Nuijten et al., 2015). An automatically generated report follows.

The scan detected 9 statistical results in APA format, of which 0 contained potentially incorrect statistical results, of which 0 may change statistical significance (alpha = .05). Potential one-tailed results were taken into account when 'one-sided', 'one-tailed', or 'directional' occurred in the text.

Note that these are not definitive results and require manual inspection to definitively assess whether results are erroneous.

Reference
Nuijten, M. B., Hartgerink, C. H. J., van Assen, M. A. L. M., Epskamp, S., & Wicherts, J. M. (2015). The prevalence of statistical reporting errors in psychology (1985-2013). Behavior Research Methods.
http://dx.doi.org/10.3758/s13428-015-0664-2

`pubpeer.com`

- Tortured phrases
- Specific phrases : *...in the era of deep learning...*
- Math (!)

⚠ LLM are hard to detect... But it is easy to be caught!

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- Prompt forgetting :
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- Specific phrases : *...in the era of deep learning...*
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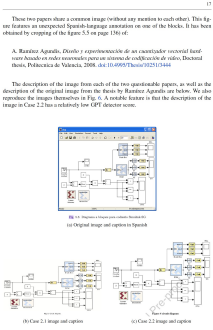

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These two papers share a common image (without any mention to each other). This figure features an unexpected Spanish-language annotation on one of the blocks. It has been obtained by cropping of the figure 5.5 on page 136) of:

A. Ramírez Agundis, *Diseño y experimentación de un cuantizador vectorial basado en redes neuronales para un sistema de codificación de vídeo*, Doctoral thesis, Politécnica de Valencia, 2008. doi:10.4995/Thesis/10251/2444.

The description of the image from each of the two questionable papers, as well as the description of the original image from the thesis by Ramírez Agundis are below. We also reproduce the images themselves in Fig. 6. A notable feature is that the description of the image in Case 2.2 has a relatively low GPT detector score.

Fig. 5 S. Dispositivo Adaptado para codificador Basado-RN

(a) Original image and caption in Spanish

(b) Case 2.1 image and caption

(c) Case 2.2 image and caption

Figure 6 Image and caption

Fig. 6 Original image from a doctoral thesis (a) and its cropped versions in Case 2.1 (b) and Case 2.2 (c).

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Description of the original image – GPT detector score: 0.02%

Como es bien sabido, Simulink es un entorno gráfico que permite el diseño y simulación de modelos usando una metodología basada en diagramas de bloques. Proporciona además bibliotecas de bloques para las tareas comunes de procesamiento en muy diversas áreas y, lugar es importante para el estudiado, crear nuevas bibliotecas e incorporar otras proporcionadas por terceras partes, en forma de ficheros binarios compilados. Por otro lado, Simulink es un paquete de software a Matlab, a través de System Generator y las del fabricante de la placa Avnet vest ampoula la FPGA. La Fig. 5.5 muestra un ejemplo que incluye bloques de los tres tipos.

Description of the image in Case 2.1 – GPT detector score: 99.98%

Fig. 5 shows as if all is considered to be completed, after the hunter and vital contamination. Tcglossaj period based on the average cycle opuesto clinical course staged the fibrilo period, step-down stage, hypoactiuary stage diuretic phase. It can be improved—an essential part of all stages of kidney contribution.

Description of the image in Case 2.2 – GPT detector score: 86.75%

Figure 4 the following stage expects so to change from Halfdood to chances. The pre-test changes can be determined utilizing the recently sketched net conditions (see area "Relative changes and chances proportions") or determined utilizing the pre test likelihood.

Additionally, we show other samples of questionable text in Cases 2.1 and 2.2, with some irregularities highlighted.

Case 2.1, related work – GPT detector score: 99.98%

Gadget embedded clinical tools are typically coupled to locate, visualize and test the disease [1]. With such a device, including bmy period and **most manifested people: metere above sex level, cardiovascular screen, screen blood glucose**, the ECG (Electro Cardio Graym, X-ray imaging, the MRI (Magnetic Resonance Imaging), CT (Computed Tomography) and PET (Positron Emission Tomography). (After the elimination of human formalities) the installation size and cost associated **with the equipment are considerably reduced. Moreover, the operator** (medical analyzer) due to the **dynamic need to operate** it is to perform the development control framework.

Case 2.2, conclusion – GPT detector score: 99.99%

Competitor's exhibition to anticipate, recommend the sequential respect of elective forward techniques. The proposed innovation is an altered form of the group of the more learning model. **Utilizing many arranged history neural organizations**, searching the principle from with the proper boundaries, and sporting the best model with at least wellness work in the initial set. **To ensure the proposed exposition's greatness,** competitors' informational execution index around the globe has been utilized.

6.3 Case 3: Circuits *Today heart rate monitor presented as something else*

▷ *Computer aided intelligent medical system and warning of breast surgery infection*, Published in volume 81 of March 2021, doi:10.1016/j.micpro.2020.103769.

Figure 2 in Case 3 contains a clear indication of its source (www.circuitstoday.com) and is most probably reused from https://todayscircuits.wordpress.com/2014/06/26/c-heart-rate

Code Generation

Getting started with a new library

- Start learning with simple examples
- Ask for Documentation

- Regarding machine learning and scikit learn library. Can you generate a python code that generate a 2 class toy example and compare linear and random forest classifer?
  - Ask the LLM to explain some part of the code !

- For instance: write a python/numpy short program that generate random point and plot them using `bokeh` so that index are displayed when the mouse passes over the point

- Generate a pink panther in html and visualize in your browser

Limitations

Bibliography

- Generate one bibliography on your favorite research topic
- Why the results won't be good?
- How may LLMs be used for this exercise?

Animals & syllabus

- Generate 10 animals the name of which count 4 syllabus

Run LLM Locally

OLlama: easy way to run locally

- LLM are huge and costly (both in computation & memory)
- ... But they have been dramatically optimized !
  - Quantization, pruning...

$\Rightarrow$ They can run locally on your machine

Simple solution: `ollama`: `https://ollama.com`

OLlama: easy way to run locally

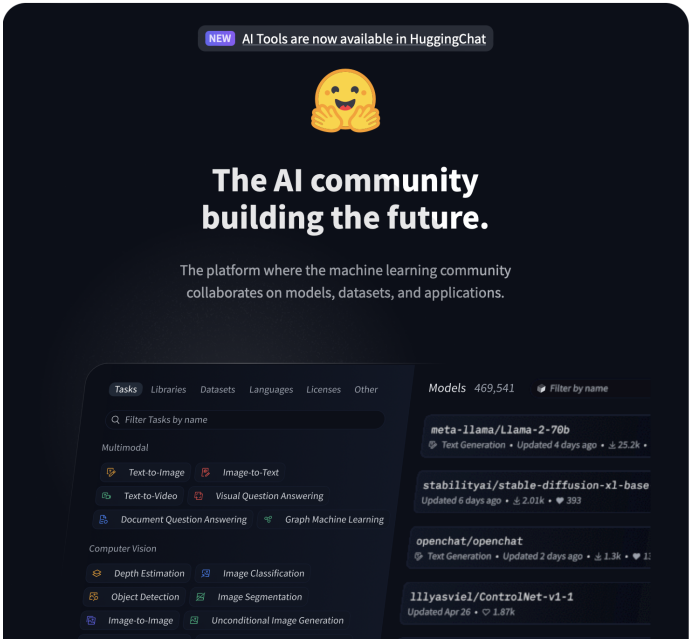
Here are some example models that can be downloaded:

Model	Parameters	Size	Download
Llama 3	8B	4.7GB	`ollama run llama3`
Llama 3	70B	40GB	`ollama run llama3:70b`
Phi 3 Mini	3.8B	2.3GB	`ollama run phi3`
Phi 3 Medium	14B	7.9GB	`ollama run phi3:medium`
Gemma 2	9B	5.5GB	`ollama run gemma2`
Gemma 2	27B	16GB	`ollama run gemma2:27b`
Mistral	7B	4.1GB	`ollama run mistral`
Moondream 2	1.4B	829MB	`ollama run moondream`
Neural Chat	7B	4.1GB	`ollama run neural-chat`
Starling	7B	4.1GB	`ollama run starling-lm`
Code Llama	7B	3.8GB	`ollama run codellama`
Llama 2 Uncensored	7B	3.8GB	`ollama run llama2-uncensored`
LLaVA	7B	4.5GB	`ollama run llava`
Solar	10.7B	6.1GB	`ollama run solar`

> Note: You should have at least 8 GB of RAM available to run the 7B models, 16 GB to run the 13B models, and 32 GB to run the 33B models.

Developments around LLM

HuggingFace



+Great tutorials
https://huggingface.co/learn/nlp-course/fr/chapter1/1

Example

```python
from transformers import pipeline
ner = pipeline("ner", grouped_entities=True)
ner("My name is Sylvain and I work at Hugging Face in Brooklyn.")
```

```
[{'entity_group': 'PER',
 'score': 0.9981694,
 'word': 'Sylvain',
 'start': 11,
 'end': 18},
{'entity_group': 'ORG',
 'score': 0.9796019,
 'word': 'Hugging Face',
 'start': 33,
 'end': 45},
{'entity_group': 'LOC',
 'score': 0.9932106,
 'word': 'Brooklyn',
 'start': 49,
 'end': 57}]
```

Example (2)

```
from transformers import pipeline

classifier = pipeline("sentiment-analysis")
classifier([
"I've been waiting for a HuggingFace course my whole life.",
"I hate this so much!"
])
```

```
[{'label': 'POSITIVE', 'score': 0.9598049521446228},
{'label': 'POSITIVE', 'score': 0.996126115322113}]
```

Example (3)

Using llama-3

```
from transformers import pipeline

classifier = pipeline("sentiment-analysis")
classifier([
"I've been waiting for a HuggingFace course my whole life.",
"I hate this so much!"
                ])
```

```
[{'label': 'POSITIVE', 'score': 0.9598049521446228},
{'label': 'POSITIVE', 'score': 0.996126115322113}]
```